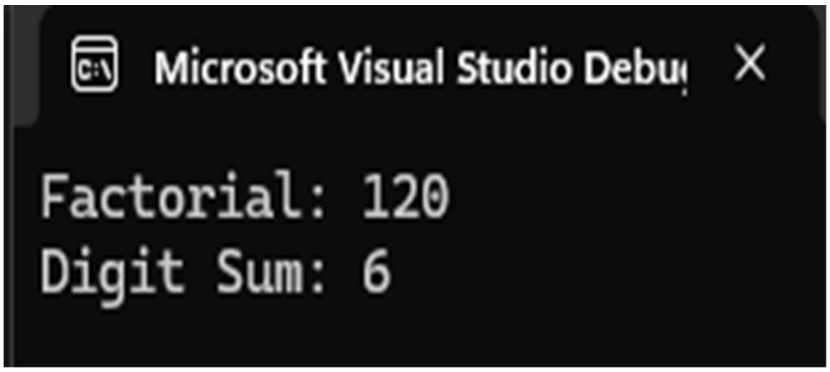
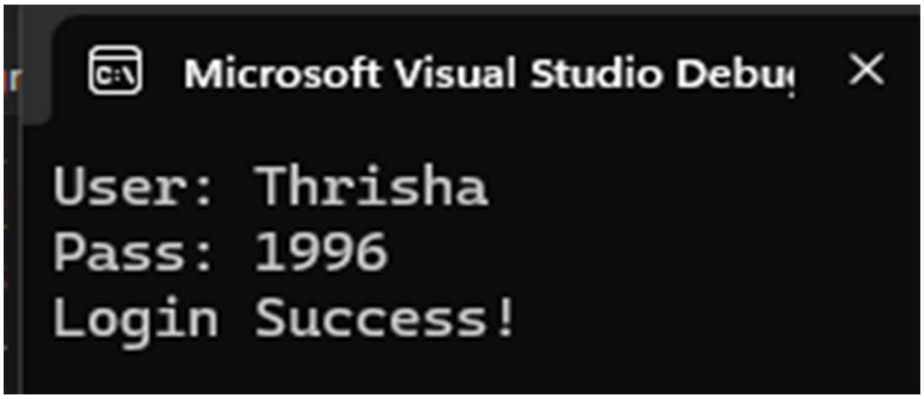
6. Program to find factorial and sum of digits using recursion.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace _6_program
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine($"Factorial: {Fact(5)}");     // 120
            Console.WriteLine($"Digit Sum: {SumDig(123)}"); // 6
        }
        // Short recursive Factorial
        static long Fact(int n) => n <= 1 ? 1 : n * Fact(n - 1);
        // Short recursive Sum of Digits
        static int SumDig(int n) => n == 0 ? 0 : (n % 10) + SumDig(n / 10);
    }
}
```



7. Program to validate a username and password using conditional logic.

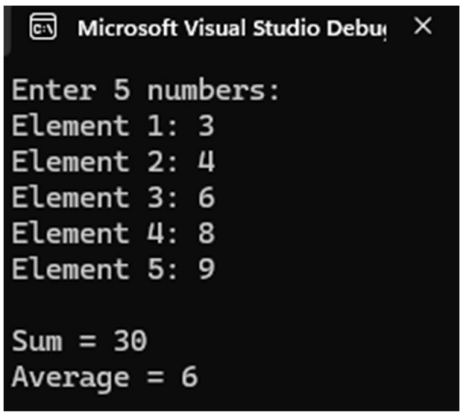
```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace _7_program
{
    internal class Program
    {
        static void Main(string[] args)
        {
            string user = "Thrisha", pass = "1996";
            Console.Write("User: "); string u = Console.ReadLine();
            Console.Write("Pass: "); string p = Console.ReadLine();
            // One-line conditional check
            string result = (u == user && p == pass) ? "Login Success!" : "Invalid Credentials";
            Console.WriteLine(result);
        }
    }
}
```



8. Program to demonstrate one-dimensional array operations (Sum , Average).

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace _9_program
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] numbers = new int[5];
            int sum = 0;
            Console.WriteLine("Enter 5 numbers:");
            // Input elements into array
            for (int i = 0; i < numbers.Length; i++)
            {
                Console.Write("Element " + (i + 1) + ": ");
                numbers[i] = int.Parse(Console.ReadLine());
            }
            // Calculate sum
            for (int i = 0; i < numbers.Length; i++)
            {
                sum += numbers[i];
            }
            // Calculate average
            double average = (double)sum / numbers.Length;
```

```
            Console.WriteLine("\nSum = " + sum);
            Console.WriteLine("Average = " + average);
        }
    }
}
```

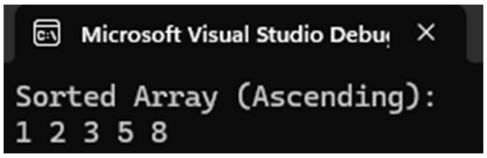


9. Program to sort elements in an array in ascending order.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace _10_program
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] arr = { 5, 2, 8, 1, 3 };
            Array.Sort(arr);

            Console.WriteLine("Sorted Array (Ascending):");
```

```
            foreach (int num in arr)
            {
                Console.Write(num + " ");
            }
        }
    }
}
```



10. Program to demonstrate string operations (length, concatenation, comparison).

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace _8_program
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter first string: ");
            string str1 = Console.ReadLine();
            Console.Write("Enter second string: ");
            string str2 = Console.ReadLine();
            // Length
```

```
        Console.WriteLine("Length of first string: " + str1.Length);
        // Concatenation
        string combined = str1 + " " + str2;
        Console.WriteLine("Concatenated String: " + combined);
        // Comparison
        if (str1.Equals(str2))
        {
            Console.WriteLine("Both strings are equal.");
        }
        else
        {
            Console.WriteLine("Strings are not equal.");
        }
    }
  }
}
```